

P V KESHAV

Mechanical Design Engineer

Kochi, Kerala | +91 7356936071 | pvkeshav.mec@gmail.com | [Portfolio](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

Mechanical Design Engineer with hands-on experience designing and analyzing medical equipment for a biotechnology device manufacturer. Skilled in CAD/CAE tools (SolidWorks, Fusion 360, ANSYS Workbench/Fluent) and CFD analysis of fluidic and clinical cartridge systems, with practical exposure to CNC machining and 3D printing for precision prototyping. Experienced in supporting full product development cycles for diagnostic and medical devices, from concept modeling through simulation-driven design validation and physical prototyping. Seeking to bring design engineering and analysis expertise to a medical device development team.

TECHNICAL SKILLS

Medical Device & Product Design: Medical equipment design, diagnostic device development, prototyping, CFD analysis, design for manufacturing (DFM), design validation

CAD / CAE Software: SolidWorks, Autodesk Fusion 360, CATIA, AutoCAD, ANSYS Workbench, ANSYS Fluent

Manufacturing & Prototyping: CNC machining, CNC lathe operation, resin 3D printing, FDM 3D printing, jig boring

Design & Visualization: Adobe Illustrator, Adobe Photoshop, Blender, Figma

Core Competencies: Teamwork, Leadership, Time Management, Problem Solving

WORK EXPERIENCE

Junior Design Engineer — UBIO Biotechnology Systems

May 2025 – Present

Kochi, Kerala

- Design and analyze medical equipment using Fusion 360, SolidWorks, and ANSYS, translating clinical and functional requirements into manufacturable mechanical designs.
- Support prototyping of medical device components through CNC machining, enabling precision manufacturing and rapid design iteration.

Mechanical Engineering Intern — UBIO Biotechnology Systems

6 Months

Kochi, Kerala

- Built expertise in advanced manufacturing methods for medical devices, including CNC machining and resin/standard 3D printing, to support device prototyping.
- Performed CFD analysis and mechanical design work using ANSYS and Fusion 360 to evaluate design performance prior to fabrication.

Mechanical Engineering Intern — HMT

1 Week

Kochi, Kerala

- Gained hands-on exposure to precision manufacturing using CNC lathe machines and jig boring machines.

Mechanical Engineering Intern — Southern Railway

1 Week

Kochi, Kerala

- Trained in maintenance and repair of diesel locomotives, working with diagnostic tools and electric/hydraulic systems.

UNIVERSITY PROJECTS

Fully Automated RNA Extraction and Analysis Device

Ongoing

Design & Analysis Lead | Team of 6

- Designing and developing an automated RNA extraction and analysis device using ANSYS, Fusion 360, CNC machining, and resin/standard 3D printing.
- Conducting design analysis to validate production readiness of a diagnostic medical device.

CFD Analysis of Clinical Cartridge

Ongoing

CFD Analyst | Team of 2

- Analyzing fluid flow within a closed clinical cartridge using ANSYS Workbench and Fluent to ensure optimal flow behavior and functional performance.

Performance Enhancement of Solar Still <i>Fabrication & Analysis Team of 4</i> Designing and fabricating a solar still integrating phase change materials (PCMs) to improve thermal efficiency and water distillation output.	Ongoing
Formula Kart Design Challenge (FKDC) <i>Design & Analysis Team of 15</i> Engineered and built a high-performance go-kart using Fusion 360, ANSYS, CNC machining, and 3D printing.	Completed

POSITIONS OF RESPONSIBILITY

Placement Cell, Govt. Model Engineering College <i>Kochi, Kerala</i> - Coordinated placement activities for 500+ students and built relationships with 100+ recruiters, increasing available placement opportunities by 30% year-over-year. - Fostered a collaborative team environment ensuring effective communication and execution of placement operations. - Member, GDC MEC 2023 — game development club of Govt. Model Engineering College.	April 2024 – April 2025
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CERTIFICATIONS & TRAINING

- “[Bridging the Gap: Developing an Employable Skilled Workforce in Material Characterisation, Testing and Analysis](#)” — jointly organized by Coventry University (UK), IIT Guwahati, and Numaligarh Refinery Limited under the British Council's Going Global Partnership Grant.
- [AutoCAD Certification](#) — CADD Campus, Autodesk.
- [Computational Fluid Dynamics Certification](#) — COGNICOR AICTE IDEA LAB.
- IC Engines Workshop — Petrichor '23, IIT Palakkad.

EDUCATION

Bachelor of Technology in Mechanical Engineering — Govt. Model Engineering College <i>Kochi, Kerala CGPA: 7.0/10</i>	May 2025
Class XII — Champion School <i>Kochi, Kerala GPA: 8.5/10</i>	April 2021
Class X — Champion School <i>Kochi, Kerala GPA: 8.4/10</i>	April 2019